

			magnitude of v_y	
2	(a)		Total momentum is constant if net external force acting on the system is zero	1 1
	(b)		$3mv + 2m(-v) = 3mv' + 2mv'$ $v' = 0.20 v$ Deduct sf mark if student leaves answer in fraction. Allow 1 sf in this case.	1 1
	(c)		<u>Total initial momentum</u> of the system is <u>non-zero</u> . <u>By principle of conservation of momentum</u> , the total momentum of the system at any point in time should not be zero. OR If both stop at the same instant, the total momentum of the system will be zero, hence it violates the principle of conservation of momentum.	1 1
	(d)		Taking rightwards as positive, By conservation of momentum, $3mv + 2m(-v) = 3m(v_T) + 2m(v_D)$ $v = 3v_T + 2v_D$ Relative speed of approach = relative speed of separation $v - (-v) = v_D - v_T$ $v_T = v_D - 2v$	1 1

			Substituting, $V = 3(V_D - 2V) + 2 V_D$ $V_D = 1.4 V$ $V_T = -0.6 V$	1 1